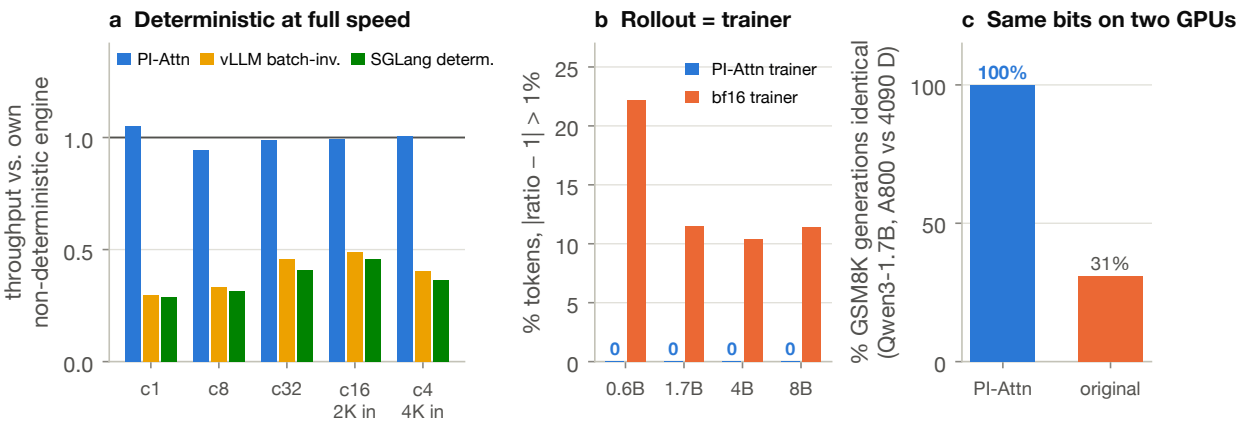
a Deterministic at full speed
PI-Attn
vLLM batch-inv.
SGLang determ.
throughput vs. own non-deterministic engine
1.0
0.5
0.0
c1
c8
c32
c16 2K in
c4 4K in
b Rollout = trainer
PI-Attn trainer
bf16 trainer
% tokens, |ratio − 1| > 1%
25
20
15
10
5
0
0
0
0
0
0.6B
1.7B
4B
8B
c Same bits on two GPUs
% GSM8K generations identical (Qwen3-1.7B, A800 vs 4090 D)
100%
100
50
0
31%
PI-Attn
original